

# Customer Churn Insights

Streamlit web dashboard for churn prediction, model comparison, explainability, and retention recommendations.

Customers

7,043

Churn rate

26.5%

Best model

Gradient Boosting

Best ROC-AUC

0.843

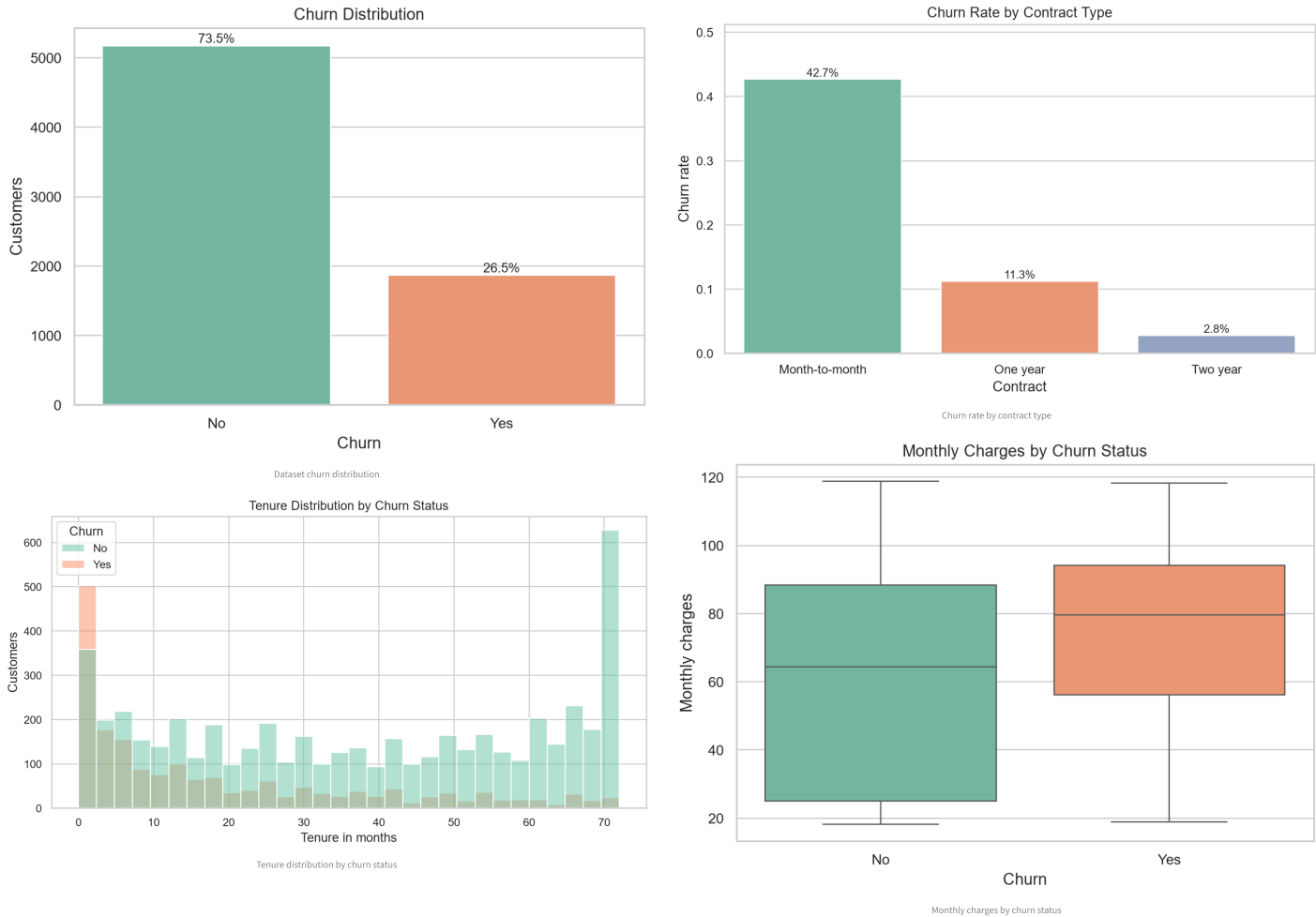
Prediction

Model Performance

Threshold Tuning

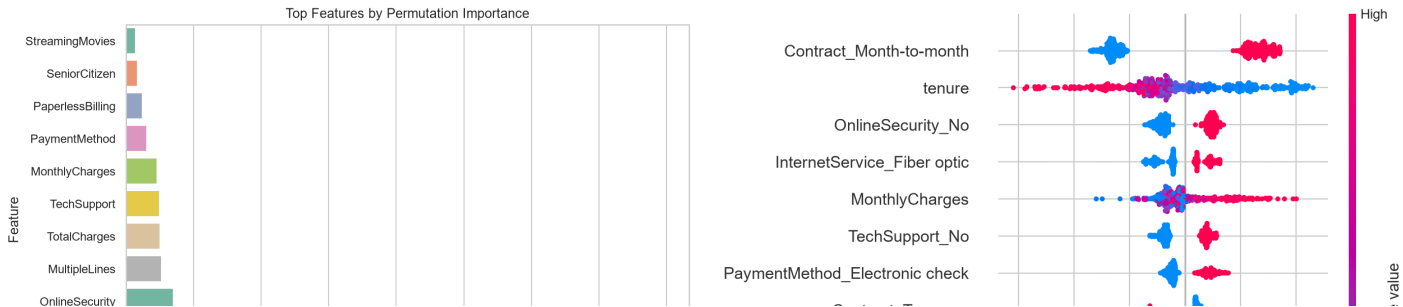
Business Insights

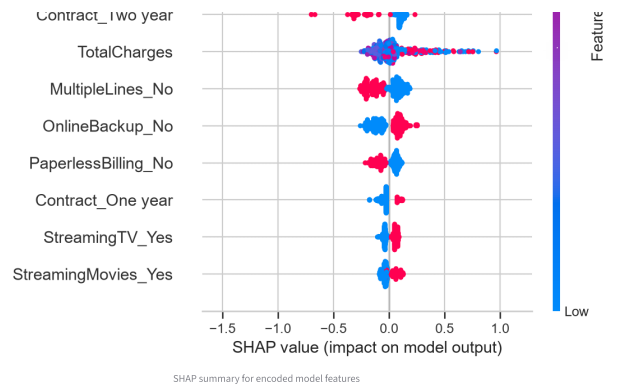
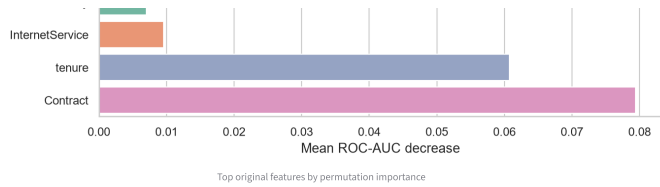
## Churn Drivers



## Explainability

|   | feature          | importance_mean | importance_std |
|---|------------------|-----------------|----------------|
| 0 | Contract         |                 | 0.07940.0053   |
| 1 | tenure           |                 | 0.06070.0016   |
| 2 | InternetService  |                 | 0.00960.0040   |
| 3 | OnlineSecurity   |                 | 0.00700.0014   |
| 4 | MultipleLines    |                 | 0.00530.0009   |
| 5 | TotalCharges     |                 | 0.00510.0018   |
| 6 | TechSupport      |                 | 0.00500.0018   |
| 7 | MonthlyCharges   |                 | 0.00460.0033   |
| 8 | PaymentMethod    |                 | 0.00310.0010   |
| 9 | PaperlessBilling |                 | 0.00240.0018   |





## Retention Actions

- Prioritize short-tenure customers with month-to-month contracts.
- Review high monthly charges for customers with elevated churn probability.
- Promote security, backup, and support services to increase product engagement.
- Tune the decision threshold before launching a campaign, because campaign cost and expected retained revenue determine the optimal cutoff.